

OpenWearables 2024: 1st International Workshop on Open Wearable Computers

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Figure 1: The 1st International Workshop on Open Wearable Computers (OpenWearables 2024) aims to create a research platform that increases the accessibility and utility of wearable technologies by focusing on the development and dissemination of open devices. (AI-generated teaser figure).

ABSTRACT

Open hardware such as Arduino are an accelerator for research in ubiquitous and wearable computing. In recent years, an increasing number of open-source wearable devices is emerging. In this workshop, we seek to create a dedicated forum and venue for publication for topics around open wearable computers. The whole day workshop includes a keynote speech, paper presentations, demo sessions, group discussions, and networking opportunities. Through our activities, we hope to create a future where wearable technologies are accessible, interoperable and impactful across applications and industries.

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CCS CONCEPTS

• **Hardware** → Emerging technologies; • **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Ubiquitous and mobile devices;

KEYWORDS

open wearables; open hardware; open-source; OSHW

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1 INTRODUCTION

In today's rapidly expanding wearable technology market, thousands of devices are available, from health trackers to augmented reality interfaces. However, a significant barrier to innovation and research in this field is the closed nature of many of these devices.

They often come with proprietary systems that are difficult to interface with or retrieve data from. Furthermore, the few open devices that are available present a steep learning curve, preventing rapid, impactful research and application development in realistic environments.

As general-purpose platform, Arduino has found wide adoption in ubiquitous and wearable computing. Extending beyond this general platform, open-source wearable devices are beginning to be more pervasive in research for example OpenBCI [2], EmotiBit [5], OpenEarable [7], and Open ExG Headphones [4].

The OpenWearables workshop aims to create a research platform that increases the accessibility and utility of wearable technologies by focusing on the development and dissemination of open devices. In recent years, communities like the Open Source Hardware Association (OSHA) have made significant strides in promoting open standards and guidelines for creating high quality, reproducible hardware [1]. The OSHA and DIN SPEC 3105 [3] have already proposed general standards for open hardware, but there is a growing need to be more specific and practical when it comes to wearable technology. Similarly, the multi-company OWear initiative [6] seeks to build a community for the collection of open data for wearable computing. In contrast to these efforts, our workshop will focus more on creating a practical exchange of wearable technology that meets the needs of researchers beyond processes and standards, and will not address open data for wearable computing.

The wearable systems should be designed for quick, easy and impactful use with a focus on:

- **Openness:** Enabling developers to create custom applications and services that communicate seamlessly with the devices.
- **Reproducibility:** Enabling anyone to build or modify the devices, ensuring that the knowledge and technology is freely available.
- **Extensibility:** Enabling easy physical integration (e.g. via hardware connectors) and software integration (e.g. via communication interfaces) with other devices and systems that encourage cross-use within the community.

2 GOAL AND OBJECTIVES

We seek to bring together people from different backgrounds that are passionate about open wearable computing. The OpenWearable workshop is driven by the following five key objectives:

- (1) **Democratize wearable technology:** We aim to make advanced wearable technologies accessible to a wider audience, including hobbyists and educators, thereby democratizing innovation in the field.
- (2) **Foster a community of open innovation:** We seek to create a forum to learn about each other's activities and create a vibrant ecosystem where developers, researchers and users can collaborate, share results and build on each other's work to advance the field.
- (3) **Enable cross-disciplinary innovation:** By lowering the barrier to entry for the use and development of wearable technologies, we aim to enable researchers and innovators from diverse fields - from healthcare to education and beyond - to use these tools in their work.

- (4) **Accelerate real-world applications:** By simplifying the development and integration process, we aim to shorten the time from concept to deployment of wearable technologies in real-world applications, making it easier to address immediate societal needs.
- (5) **Make interoperability easier:** Establish useful, concrete interfaces that are needed by the wearable community to ensure interoperability and ease of use across devices and platforms.

Beyond these five core principles, the OpenWearables workshop is not only a meeting place for current technologies, but also a launch pad for future innovations. Through collaboration and open exchange of ideas, we aim to set new standards for wearable technologies to be compatible. This way barriers to entry are lowered and a community of passionate individuals and organizations can work together towards common goals.

3 BACKGROUND OF THE ORGANIZERS

Tobias Röddiger

Research Group Leader, Karlsruhe Institute of Technology, Germany
Tobias is a group leader for wearable systems at the Karlsruhe Institute of Technology in Germany. He is the developer of OpenEarable, an open-source Arduino platform for ear-based sensing applications. He regularly serves as reviewer for different journals and conferences (e.g. IMWUT, CHI) and has helped organize various events (e.g. UbiComp, MuC).

Michael Beigl

Professor, Karlsruhe Institute of Technology, Germany
Michael Beigl is a researcher and practitioner in the field of Ubiquitous Computing and Wearable Computing. He has served as Program and General Co-Chair of UbiComp and ISWC 2016, 2018, 2021, and co-organized several workshops on UbiComp/ISWC.

Kristof Van Laerhoven

Professor, University of Siegen, Germany
Kristof is a professor for Ubiquitous Computing and interested in research into wearable and embedded sensing systems, as well as activity and affect recognition. He has co-chaired UbiComp 2023 and chaired till recently the ISWC Steering Committee, and currently is an Editor for the IMWUT journal.

Katia Vega

Associate Professor, University of California, Davis
Katia Vega is an Associate Professor at UC Davis, heading the Interactive Organisms Lab. Her research interests involve Beauty Technology, embedding electronics in cosmetics, Body Modification Technologies, integrating electrochemical and colorimetric biosensors into jewelry, tattoos and dental orthodontics for sensing biofluids, and Growable Interfaces, creating sustainable interfaces with bio-based materials. She has served as a program co-chair and co-chair of the Design Exhibition at ISWC.

4 EXPECTED ATTENDANCE

The OpenWearables workshop aims to engage a group of 20-40 in-person participants who embody the spirit of openness and diversity of expertise, ranging from hardware enthusiasts to software developers, designers, artists, and systems builders that can contribute to the field of Wearable Computing. The wide range of backgrounds will offer a basis for future collaboration. We will also livestream the event for participants that would like to join online.

We will recruit attendees from the community via the open-wearables.org website that we will share on social media, via e-mail, and by personally approaching various researchers within our network who are working on open wearable computers. Furthermore, we are inviting anyone attending Ubicomp/ISWC to participate in all sessions.

5 AGENDA

The full day workshop will be divided into four main parts: a keynote, paper presentations, demo sessions and a community discussion session.

5.1 Keynote

We will open the workshop with a keynote by a speaker that is relevant to the open wearables community. The goal of this talk will be to inspire the attendees for further research and showcase a success story of open wearable computing.

5.2 Papers

This session includes presentations based on accepted and invited papers. It will serve as a platform for sharing recent developments, findings and innovations in the field of open technology that can be or are used in the wearable computing domain. We invite papers including but not limited to the following topics:

- open-source designs of wearable devices
- applications and evaluation of open-source wearables
- software that supports designing, developing, or building open-source wearables
- frameworks simplify the interaction between open-source wearables

Submitted papers should be short, not exceeding 4 pages (excluding references and additional material). Papers should be concise and clear when presenting technical details (e.g., bullet lists, figures, etc.). We expect that most papers will present open hardware, software, or systems and their intended use and benefits. A significant part of the paper should be dedicated to the aspect of how to use, build and interface the presented approach. Complexity levels should indicate what skills and technical settings are required to use the system. At least one application example should demonstrate the use and impact. We expect that hard- and software presented at the workshop is open source. Papers that present a novel piece of hardware or software will have to include all relevant files as supplemental material in their submission.

5.2.1 Review Process. For paper review, we will set up a program committee. Each paper will be reviewed by two different members of the committee. Following the reviews, the organizers together

with the PC members will discuss a meta review for the final selection. The meta review will summarize the reviews and give recommendations for the final submission. Supplemental material will be lightly reviewed.

5.2.2 Publication. We intend to have all papers published in the “ACM Digital Library and Adjunct Proceedings”. All papers have to follow the requirements of ACM double-column format (UbiComp/ISWC 2024 Proceedings Formats). Papers, presentations, demos videos and supplemental materials will also published on open-wearables.org.

5.3 Demo Sessions

Participants are encouraged to demonstrate or showcase their open source wearable projects. This hands-on session aims to showcase the practical applications of open wearables and encourage direct feedback and engagement from attendees. This session is open to all workshop participants: Bring your own device and/or software! Sessions will be limited in time (around 5 minutes) to give each participant the opportunity to showcase their work in progress and also to make connections between projects. Participants can opt-in to have their demo shared on open-wearables.org.

5.4 Group Discussions: Towards General-Purpose Open Wearables

This session will focus on collective brainstorming and strategizing to address common challenges - also between sub-groups - and goals within the open wearables community. Key discussion points will include but are not limited to:

- Improving collaboration within the community to unify efforts and create synergies.
- Tools to facilitate the creation of open software and hardware for wearables.
- Setup processes for wearables to increase accessibility and user adoption.
- Strategies for publishing in A+ venues to gain wider recognition and impact.

Depending on the number of participants, the format of this coffee break discussion can vary. If there is a smaller group of participants, we may have a single discussion. Alternatively, if there are more people, we will break into smaller groups to facilitate more in-depth conversations and ensure that everyone has the opportunity to participate, followed by a discussion with all participants. We aim to disseminate the findings into a short position paper that will be published as part of the workshop proceedings. If possible, we also want to publish a small magazine article in *Communications of the ACM* that summarizes the key findings of the workshop.

5.5 Networking

We intend to have a coffee and lunch break which create opportunities for participants to get to know each other better. They can discuss potential collaborations within the open wearable community and identify application partners for their open wearable devices. This will initiate the formation of an international open wearable community.

5.6 Awards

As part of the workshop we will award two prizes.

- **Best Paper Award:** The prize will be awarded to the best paper, based on the paper itself and the presentation. This prize will be selected from the organizing committee.
- **Best Demo Award:** This prize will be awarded to the best demo and will be based on votes from the workshop participants.

Both awards will serve as a testament to the outstanding work being done in the field of open wearables and should inspire future research and development in this area.

5.7 Preliminary Schedule

We plan to run a full day workshop with the following schedule:

Table 1: Preliminary schedule for the OpenWearables 2024 workshop in Melbourne.

Time	Session Name
09:00 - 09:45	Introduction and Keynote
09:45 - 10:00	Coffee Break
10:00 - 12:00	Paper Presentations
12:00 - 13:00	Lunch Break
13:00 - 14:00	Demo Session
14:00 - 15:45	Discussion and Coffee
15:45 - 16:00	Wrap-Up and Awards

6 POST WORKSHOP FOLLOW-UP

All materials, including presentations, papers and demo videos, will be made available on open-wearables.org (and / or Github), serving as a comprehensive resource for the community. We encourage continued collaboration and sharing of insights through the forums and contribution opportunities on the website and via social media channels as well. To understand how we can improve the workshop in its future editions, we will survey the participants. We will also provide updates on future events and initiatives aimed at advancing open wearable technologies, and seek to coordinate community efforts. The workshop is planned to be repeated every year.

REFERENCES

- [1] 2024. Open Source Hardware Association (OSHWA). <https://www.oshwa.org/>. Accessed: 2024-03-22.
- [2] 2024. OpenBCI. <https://openbci.com/>. Accessed: 2024-03-22.
- [3] Deutsche Industrienorm e.V. 2023. DIN SPEC 3105. <https://gitlab.com/OSGermany/OHS-3105>.
- [4] Michael Thomas Knierim, Daniel Puhl, Gabriel Ivucic, and Tobias Röddiger. 2023. OpenBCI+ 3D-Printed Headphones= Open ExG Headphones=An Open-Source Research Platform for Biopotential Earable Applications. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–7.

- [5] Sean M Montgomery, Nitin Nair, Phoebe Chen, and Suzanne Dikker. 2023. Introducing EmotiBit, an open-source multi-modal sensor for measuring research-grade physiological signals. *Science Talks* 6 (2023), 100181.
- [6] Open Wearables Initiative. 2024. OWeAR - Open Wearables Initiative. <https://www.owear.org/>. Accessed: 2024-03-20.
- [7] Tobias Röddiger, Tobias King, Dylan Ray Roodt, Christopher Clarke, and Michael Beigl. 2023. OpenEarable: Open Hardware Earable Sensing Platform. In *Proceedings of the 1st International Workshop on Earable Computing* (Cambridge, United Kingdom) (*EarComp'22*). Association for Computing Machinery, New York, NY, USA, 29–34. <https://doi.org/10.1145/3544793.3563415>

A CALL FOR PAPERS: OPENWEARABLES 2024 – 1ST INTERNATIONAL WORKSHOP ON OPEN WEARABLE COMPUTERS

The OpenWearables 2024 workshop aims to address the challenges and opportunities in the field of open source wearable technology. We invite submissions from researchers, developers and innovators on topics such as open source designs of wearable devices, applications and evaluations of open source wearables, software that supports the design and development of open wearables, and frameworks. Submissions should be concise, limited to a maximum of 4 pages excluding references, and demonstrate the use, build and interface processes of open hardware, software or systems. In addition, it is expected that authors submit "open-source materials" which are, for example, design files, 3D-printing files, schematics or any other materials that are required to reproduce the wearable device. Papers should use pictures, graphs and functional diagrams as often as possible in the explanation of the work. An essential requirement is that the projects presented adhere to open source principles. Papers will be selected based on adherence to these principles and the clarity of the paper. During the workshop, authors will be required to present their research paper and also provide a demonstration of their open wearable work to showcase the practical applications and potential impact of their research.

The workshop will feature a mix of keynote speeches, paper presentations, demo sessions, and group discussions, providing a platform for participants to showcase their work, share insights, and foster collaboration within the open wearables community. We particularly encourage demonstrations of open source wearable projects during the hands-on demo sessions, in addition to the paper.

All accepted papers will be considered for inclusion in a special position paper summarising the results of the workshop, which will be published in the proceedings. We will also make all workshop materials available on open-wearables.org and GitHub, creating a lasting resource for the community.

Join us at OpenWearables 2024 to help democratise wearable technology, accelerate innovation and establish standards for open wearables. Together, we can create a future where wearable technologies are accessible, interoperable and impactful across applications and industries.